

A CHARACTERIZATION OF A MULTIVARIATE DISTRIBUTION

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1. Introduction

Seshadri and Patil (1964) derived and applied a sufficient condition for the unique existence of a bivariate distribution. In this paper a similar approach is used to derive a sufficient condition for the unique existence of a multivariate distribution. By using logical extensions in notations, the theorems are generalised to include random matrices. The theorems are applied to derive a few multivariate distributions. Moment generating functions (m.g.f.) are also derived.

2. Random Variables

Let X_1 and X_2 be two random variables which may be continuous, discrete or mixed (Hald (1968)). Let $f(x_1|x_2)$ be the conditional probability density function (p.d.f.) of X_1 on X_2 , $f_1(x_1)$ the marginal p.d.f. of X_1 and $f_2(x_2)$ the marginal p.d.f. of X_2 . Seshadri and Patil (1964) proved the following theorem:

Theorem 2.1

Given $f_1(x_1)$ and $f(x_1|x_2)$, a sufficient condition for $f_2(x_2)$, and hence for $f(x_1, x_2)$ to be unique is that the conditional p.d.f. of X_1 on X_2 is of the exponential form.

$$f(x_1|x_2) = \exp [x_2 g(x_1) + h(x_1) + t(x_2)]$$

where an interval is contained in the range of $g(x_1)$.

The following theorem will now be proved:

Theorem 2.2

Given (i) $f_{1,k-1}(x_1, x_2, \dots, x_{k-1})$, the marginal p.d.f. of $X_1, X_2, \dots, X_{k-1}$, and (ii) $f(x_1, \dots, x_{k-1}|x_k)$, the conditional p.d.f. of $X_1, \dots, X_{k-1}$ on X_k , then a sufficient condition for $f_k(x_k)$, and hence for $f(x_1, x_2, \dots, x_k)$ to be unique is that the conditional p.d.f. is of the form

$$(2.1) \quad f(x_1, \dots, x_{k-1}|x_k) = \exp [s(x_k)g(x_1, x_2, \dots, x_{k-1}) + h(x_1, x_2, \dots, x_{k-1}) + t(x_k)]$$

where $g(x_1, x_2, \dots, x_{k-1})$ and $h(x_1, x_2, \dots, x_{k-1})$ are functions of $x_1, x_2, \dots, x_{k-1}$ and $t(x_k)$ and $s(x_k)$ are functions of x_k

Proof

For any p.d.f. it follows that

$$f(x_1, \dots, x_{k-1} | x_k) | f_k(x_k) = f(x_1, \dots, x_k)$$

and

$$(2.2) \quad \int f(x_1, \dots, x_{k-1} | x_k) f_k(x_k) dx_k = f_{1,k-1}(x_1, \dots, x_{k-1})$$

where

$$f_{1,k-1}(x_1, \dots, x_{k-1}) = \int f(x_1, \dots, x_k) dx_k$$

From (2.1) and (2.2) it follows that

$$(2.3) \quad \int \exp[s(x_k)g(x_1, \dots, x_{k-1})] \phi(x_k) dx_k = \psi(x_1, \dots, x_{k-1})$$

where

$$\phi(x_k) = f_k(x_k) \exp[t(x_k)]$$

and

$$\psi(x_1, \dots, x_{k-1}) = f_{1,k-1}(x_1, \dots, x_{k-1}) \exp[-h(x_1, \dots, x_{k-1})]$$

For given $f_{1,k-1}(x_1, \dots, x_{k-1})$ and $f(x_1, \dots, x_{k-1} | x_k)$ in form (2.1), $\psi(x_1, \dots, x_{k-1})$ is known and by the uniqueness of the generalised inverse Laplace transform (Widder (1941), p. 37) (Kendall and Stuart (1961), p. 190), $\phi(x_k)$ and therefore $f_k(x_k)$ can be determined uniquely under certain general restricting conditions on $s(x_k)$ and $\phi(x_k)$. This proves the theorem.

Example 2.1

Given the p.d.f. of the Dirichlet distribution of the second kind (Troskie (1967)) viz.

$$f_{1,k-1}(x_1, \dots, x_{k-1}) = \frac{\Gamma(\sum_{i=1}^{k-1} n_i + m)}{\pi^{\frac{k-1}{2}} \Gamma(n_1) \Gamma(m)} \frac{k-1}{i=1} x_i^{n_i-1} (1 + \sum_{i=1}^{k-1} x_i)^{-(\sum_{i=1}^{k-1} n_i + m)},$$

$$x_i > 0, 1 - \sum_{i=1}^{k-1} x_i > 0,$$

and the conditional p.d.f. of $X_1, \dots, X_{k-1}$ given $X_k = x_k$ as

$$f(x_1, \dots, x_{k-1} | x_k) = \frac{1}{\pi^{\frac{k-1}{2}} \Gamma(n_1)} \exp[-x_k \sum_{i=1}^{k-1} x_i] \frac{k-1}{i=1} x_i^{n_i-1} x_k^{\sum_{i=1}^{k-1} n_i}, x_i > 0,$$

it follows by applying theorem 2.1 and from (2.3) that

$$\int_0^\infty \exp[-x_k \sum_{i=1}^{k-1} x_i] \phi(x_k) dx_k = \frac{\Gamma(\sum_{i=1}^{k-1} n_i + m)}{\Gamma(m)} (1 + \sum_{i=1}^{k-1} x_i)^{-(\sum_{i=1}^{k-1} n_i + m)}$$

and by the uniqueness of the inverse Laplace transform (Erdelyi, et al (1954), p. 238) that,

$$\phi(x_k) = \frac{1}{\Gamma(m)} \exp[-x_k] x_k^{\sum_{i=1}^{k-1} n_i + m - 1}$$

and therefore

$$f(x_k) = \frac{1}{\Gamma(m)} \exp[-x_k] x_k^{m-1}, \quad x_k > 0.$$

The joint distribution of $X_1, \dots, X_k$ is then given by

$$(2.4) \quad f(x_1, \dots, x_k) = \frac{1}{\prod_{i=1}^{k-1} \Gamma(n_i) \Gamma(m)} \exp[-x_k(1 + \sum_{i=1}^{k-1} x_i)] \cdot x_k^{\sum_{i=1}^{k-1} n_i + m - 1} \prod_{i=1}^{k-1} x_i^{n_i - 1}, \quad x_i > 0, 1 - \sum_{i=1}^{k-1} x_i > 0.$$

which is now called a joint Dirichlet-gamma distribution.

The next theorem which is similar to theorem 2.2 deals with cases where the moment generating function (m.g.f.) is known.

Theorem 2.3

Given $M(a_1, \dots, a_{k-1})$ the m.g.f. of $f_{1, k-1}(x_1, \dots, x_{k-1})$ and $M(a_1, \dots, a_{k-1}; x_k)$ the m.g.f. of $f(x_1, \dots, x_{k-1} | x_k)$. A sufficient condition for $f(x_k)$ to be unique is that the m.g.f. of $f(x_1, \dots, x_{k-1} | x_k)$ is of the form.

$$(2.6) \quad M(a_1, \dots, a_{k-1}; x_k) = \exp[s(x_k) g(a_1, \dots, a_{k-1}) + h(a_1, \dots, a_{k-1}) + t(x_k)]$$

where $g(a_1, \dots, a_{k-1})$ and $h(a_1, \dots, a_{k-1})$ are functions of $a_1, \dots, a_{k-1}$ and $t(x_k)$ and $s(x_k)$ are functions of x_k . The $a_i, i = 1, \dots, k$, may be complex.

Proof

It is well known that

$$\int M(a_1, \dots, a_{k-1}; x_k) f_k(x_k) dx_k = M(a_1, \dots, a_{k-1})$$

which, because of (2.6), reduces to

$$(2.7) \quad \int \exp[s(x_k) g(a_1, \dots, a_{k-1})] \phi(x_k) dx_k = \psi(a_1, \dots, a_{k-1})$$

where $\phi(x_k) = f_k(x_k) \exp[t(x_k)]$ and

$$\psi(a_1, \dots, a_{k-1}) = M(a_1, \dots, a_{k-1}) \exp[-h(a_1, \dots, a_{k-1})]$$

By using a generalised inverse Laplace transform (Widder (1941) ch. VII), $\phi(x_k)$ and thus $f_k(x_k)$ follows uniquely under certain general restricting conditions on $s(x_k)$ and $\phi(x_k)$. This proves the theorem.

Consider the following example.

Example 2.2

We have

$$M(a_1) = (1-2a_1)^{-1/2n}$$

the m.g.f. of a χ^2 -distribution and

$$M(a_1, x_2) = \exp \left[a_1 \frac{x_2^2}{n} \right] (1-2a_1)^{-1/2(n-1)}$$

in form (2.6).

By applying theorem 2.2 it follows from (2.7) for $t(x_2) = 1$ that

$$\int_0^\infty \exp \left[a_1 \frac{x_2^2}{n} \right] f(x_2) dx_2 = (1-2a_1)^{-1/2}$$

Using the inverse Laplace transform (Erdelyi, et al (1954) p. 238) it follows that

$$f(x_2) = \frac{1}{\sqrt{2\pi n}} \exp \left[-\frac{1}{2} \frac{x_2^2}{n} \right]$$

and by Steyn and Roux (1970)

$$\begin{aligned} M(a_1, a_2) &= \int f(x_2) \exp [a_2 x_2] M(a_1; x_2) dx_2 \\ &= \exp \left[-\frac{1}{2} \frac{na_2^2}{(1-2a_1)} \right] \cdot (1-2a_1)^{-1/2n} \end{aligned}$$

This is known as the m.g.f. of the joint normal- χ^2 distribution and has been studied by Oliver (1969), p. 19.

As a further extension of theorem 2.3, random vectors may be substituted for the random variables and as an application the characterization of the well-known multivariate normal distribution follows clearly.

Example 2.3

Given (i) the m.g.f. of $X^{(1)}$: $k_1 x_1$ (Anderson (1958)) as

$$M(a^{(1)}) = \exp \left[\frac{1}{2} a^{(1)'} \Sigma_{11} a^{(1)} \right]$$

and (ii) the m.g.f. of $X^{(1)}$ given $X^{(2)} = x^{(2)}$ as

$$M(a^{(1)}; x^{(2)}) = \exp \left[x^{(2)'} \Sigma_{22}^{-1} \Sigma_{21} a^{(1)} + \frac{1}{2} a^{(1)'} \Sigma_{11.2} a^{(1)} \right]$$

which is in the form (2.6) for vectors, it follows from (2.7) that

$$\int \exp[\mathbf{x}^{(2)'} \Sigma_{22}^{-1} \Sigma_{21} \mathbf{a}^{(1)}] f(\mathbf{x}^{(2)}) d\mathbf{x}^{(2)} = \exp[\frac{1}{2} \mathbf{a}^{(1)'} \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} \mathbf{a}^{(1)}]$$

and by using the integral

$$\frac{1}{|\Sigma_{22}|^{1/2} (2\pi)^{1/2} k_2} \int \exp[-\frac{1}{2} (\mathbf{x}^{(2)} - \Sigma_{22} \Sigma_{21} \mathbf{a}^{(1)})' \Sigma_{22}^{-1} (\mathbf{x}^{(2)} - \Sigma_{22} \Sigma_{21} \mathbf{a}^{(1)})] d\mathbf{x}^{(2)} = 1$$

it follows that,

$$f(\mathbf{x}^{(2)}) = \frac{1}{|\Sigma_{22}|^{1/2} (2\pi)^{1/2} k_2} \exp[-\frac{1}{2} \mathbf{x}^{(2)'} \Sigma_{22}^{-1} \mathbf{x}^{(2)}]$$

where $\mathbf{x}^{(2)} : k_2 \times 1$ is a vector.

From Steyn and Roux (1970) follows

$$\begin{aligned} M(\mathbf{a}^{(1)}, \mathbf{a}^{(2)}) &= \int f(\mathbf{x}^{(2)}) \exp[\mathbf{a}^{(2)'} \mathbf{x}^{(2)}] M(\mathbf{a}^{(1)}; \mathbf{x}^{(2)}) d\mathbf{x}^{(2)} \\ &= \exp[\frac{1}{2} (\mathbf{a}^{(1)'} \Sigma_{11} \mathbf{a}^{(1)} + 2 \mathbf{a}^{(1)'} \Sigma_{12} \mathbf{a}^{(2)} + \mathbf{a}^{(2)'} \Sigma_{22} \mathbf{a}^{(2)})] \end{aligned}$$

3. Random matrices

Consider k random matrices $X_{(1)}, \dots, X_{(k)}$ with $f(\mathbf{x}_{(1)}, \dots, \mathbf{x}_{(k-1)} | \mathbf{x}_{(k)})$ the conditional p.d.f. of $X_{(1)}, \dots, X_{(k-1)}$ on $X_{(k)}$, $f_{1,k-1}(\mathbf{x}_{(1)}, \dots, \mathbf{x}_{(k-1)})$ the marginal p.d.f. of $X_{(1)}, \dots, X_{(k-1)}$ and $f_k(\mathbf{x}_{(k)})$ the marginal p.d.f. of $X_{(k)}$.

A theorem similar to theorem 2.2 will now be proved for random matrices.

Theorem 3.1

Given $f_{1,k-1}(\mathbf{x}_{(1)}, \dots, \mathbf{x}_{(k-1)})$ and $f(\mathbf{x}_{(1)}, \dots, \mathbf{x}_{(k-1)} | \mathbf{x}_{(k)})$, a sufficient condition for $f_k(\mathbf{x}_{(k)})$ and hence for $f(\mathbf{x}_{(1)}, \dots, \mathbf{x}_{(k)})$ to be unique is that the conditional p.d.f. is of the form

$$\begin{aligned} (3.1) \quad f(\mathbf{x}_{(1)}, \dots, \mathbf{x}_{(k-1)} | \mathbf{x}_{(k)}) &= \exp[\text{tr}[\mathbf{x}_{(k)}' g(\mathbf{x}_{(1)}, \dots, \mathbf{x}_{(k-1)}) \\ &\quad + h(\mathbf{x}_{(1)}, \dots, \mathbf{x}_{(k-1)}) + t(\mathbf{x}_{(k)})]] \end{aligned}$$

where $g(\mathbf{x}_{(1)}, \dots, \mathbf{x}_{(k-1)})'$ and $h(\mathbf{x}_{(1)}, \dots, \mathbf{x}_{(k-1)})$ are functions of $\mathbf{x}_{(1)}, \dots, \mathbf{x}_{(k-1)}$ and $t(\mathbf{x}_{(k)})$ a function of $\mathbf{x}_{(k)}$ and the multiplication of matrices holds.

Proof

It is known that

$$\int f(\mathbf{x}_{(1)}, \dots, \mathbf{x}_{(k-1)} | \mathbf{x}_{(k)}) f_k(\mathbf{x}_{(k)}) d\mathbf{x}_{(k)} = f_{1,k-1}(\mathbf{x}_{(1)}, \dots, \mathbf{x}_{(k-1)})$$

which, because of (3.1), reduces to

$$(3.2) \quad \int \exp [\operatorname{tr} [x'_{(k)} g(x_{(1)}, \dots, x_{(k-1)})]] \phi(x_{(k)}) dx_{(k)} \\ = \psi(x_{(1)}, \dots, x_{(k-1)})$$

where

$$\phi(x_{(k)}) = f_k(x_{(k)}) \exp [\operatorname{tr} t(x_{(k)})]$$

and

$$\psi(x_{(1)}, \dots, x_{(k-1)}) = f_{1,k-1}(x_{(1)}, \dots, x_{(k-1)}) \cdot \exp [-\operatorname{tr} h(x_{(1)}, \dots, x_{(k-1)})]$$

Now, for given $f_{1,k-1}(x_{(1)}, \dots, x_{(k-1)})$ and $f(x_{(1)}, \dots, x_{(k-1)} | x_{(k)})$, $\psi(x_{(1)}, \dots, x_{(k-1)})$ is known and by the uniqueness of the inverse Laplace transform (Constantine (1963)), $\phi(x_{(k)})$ and therefore $f_k(x_{(k)})$ can be determined uniquely. This proves the theorem. Theorem 3.1 also holds when a function of $x_{(k)}$ is substituted for $x_{(k)}$ under certain conditions.

Example 3.1

Let

$$f_1(x_{(1)}) = \frac{\Gamma_p(n+m)}{\Gamma_p(n) \Gamma_p(m)} |x_{(1)}|^{n-1/2(p+1)} |I + x_{(1)}|^{-(n+m)}, \\ 0 < x_{(1)} < \infty,$$

be a multivariate beta distribution of the second kind as given by Olkin and Rubin (1964) and let

$$f(x_{(1)} | x_{(2)}) = \frac{1}{\Gamma_p(n)} \exp [-\operatorname{tr} x_{(1)} x_{(2)}] |x_{(2)}|^n |x_{(1)}|^{n-1/2(p+1)}, \\ x_{(2)} > 0,$$

where $x_{(2)} > 0$ means that $x_{(2)}$ is positive definite.

By applying theorem 3.1 it follows from (3.2) that

$$\int \exp [-\operatorname{tr} x_{(1)} x_{(2)}] \phi(x_{(2)}) dx_{(2)} = \frac{\Gamma_p(n+m)}{\Gamma_p(m)} |I + x_{(1)}|^{-(n+m)}$$

and by the uniqueness of the inverse Laplace transform (Constantine (1963)) that,

$$\phi(x_{(2)}) = \frac{1}{\Gamma_p(m)} \exp [-\operatorname{tr} x_{(2)}] |x_{(2)}|^{n+m-1/2(p+1)}$$

and therefore

$$f_2(x_{(2)}) = \frac{1}{\Gamma_p(m)} \exp [-\operatorname{tr} x_{(2)}] |x_{(2)}|^{m-1/2(p+1)}, x_{(2)} > 0$$

The joint distribution of $X_{(1)}$ and $X_{(2)}$ is given by

$$(3.3) \quad f(x_{(1)}, x_{(2)}) = \frac{1}{\Gamma_p(n) \Gamma_p(m)} \exp[-\text{tr } x_{(2)}(I + x_{(1)})] \cdot \\ \cdot |x_{(2)}|^{n+m-\frac{1}{2}(p+1)} \cdot |x_{(1)}|^{n-\frac{1}{2}(p+1)}, x_{(1)} > 0, x_{(2)} > 0.$$

In example 2.1 replace the random variables with random matrices and by using the same arguments it can be shown that

$$f(x_{(k)}) = \frac{1}{\Gamma_p(m)} \exp[-\text{tr } x_{(k)}] |x_{(k)}|^{m-\frac{1}{2}(p+1)}, x_{(k)} > 0$$

Thus follows the joint distribution.

A theorem similar to theorem 2.3 will now be stated for random matrices.

Theorem 3.2

Given $M(a_{(1)}, \dots, a_{(k-1)})$, the m.g.f. of $f_{1,k-1}(x_{(1)}, \dots, x_{(k-1)})$ and $M(a_{(1)}, \dots, a_{(k-1)}; x_{(k)})$, the m.g.f. of $f(x_{(1)}, \dots, x_{(k-1)} | x_{(k)})$. A sufficient condition for $f(x_{(k)})$ to be unique is that the m.g.f. of $f(x_{(1)}, \dots, x_{(k-1)} | x_{(k)})$ is of the form

$$M(a_{(1)}, \dots, a_{(k-1)}; x_{(k)}) = \exp[\text{tr}[s(x_{(k)})' g(a_{(1)}, \dots, a_{(k-1)}) \\ + h(a_{(1)}, \dots, a_{(k-1)}) + t(x_{(k)})]]$$

where $g(a_{(1)}, \dots, a_{(k-1)})$ and $h(a_{(1)}, \dots, a_{(k-1)})$ are functions of $a_{(1)}, \dots, a_{(k-1)}$ and $s(x_{(k)})$ a function of $x_{(k)}$.

The proof of this theorem for random matrices is similar to the proof of theorem 2.3.

Example 3.2

Given the m.g.f. of the random matrices $X_{(11)}: p \times p$, $X_{(12)}: p \times p$ and $X_{(21)}: p \times p$ viz.

$$M(a_{(11)}, a_{(12)}, a_{(21)}) = |I - 2a|^{-\frac{1}{2}n} \text{ where } a = \begin{pmatrix} a_{(11)} & a_{(12)} \\ a_{(21)} & 0 \end{pmatrix} \\ = [|I - 2a_{(11)}| |I - 4a_{(21)}(I - 2a_{(11)})^{-1}a_{(12)}|]^{-\frac{1}{2}n}$$

and the m.g.f. of the conditional p.d.f. of $X_{(11)}$, $X_{(12)}$, and $X_{(21)}$ given $X_{(22)} = x_{(22)}$ as

$$M(a_{(11)}, a_{(12)}, a_{(21)}; x_{(22)}) = |I - 2a_{(11)}|^{-\frac{1}{2}n} \\ \cdot \exp[2 \text{tr } x_{(22)} a_{(21)} (I - 2a_{(11)})^{-1} a_{(12)}],$$

it follows by applying theorem 3.2 for $t(x_{(2)}) = 1$ that

$$X_{(22)} \int_0^\infty \exp[2 \operatorname{tr} x_{(22)} a_{(21)} (I - 2a_{(11)})^{-1} a_{(12)}] f(x_{(22)}) dx_{(22)} \\ = |I - 4a_{(21)}(I - 2a_{(11)})^{-1} a_{(12)}|^{-\frac{1}{2}n}$$

and therefore (Constantine (1963))

$$f(x_{(22)}) = \frac{2^{-\frac{1}{2}np}}{\Gamma_p(\frac{1}{2}n)} \exp[-\operatorname{tr} \frac{1}{2}x_{(22)}] |x_{(22)}|^{-\frac{1}{2}(n-p-1)}$$

From Steyn and Roux (1970) follows

$$M(a_{(11)}, a_{(12)}, a_{(21)}, a_{(22)}) = |I - 2a|^{-\frac{1}{2}n}, \text{ where } a = \begin{pmatrix} a_{(11)} & a_{(12)} \\ a_{(21)} & a_{(22)} \end{pmatrix},$$

the well-known m.g.f. of the Wishart matrix

$$X = \begin{pmatrix} X_{(11)} & X_{(12)} \\ X_{(21)} & X_{(22)} \end{pmatrix}$$

4. The moment generating functions

We consider here the m.g.f.'s of the distributions characterized in the previous paragraphs.

The m.g.f. of (2.4) is defined by

$$M(a_1, \dots, a_k) = \int \dots \int \exp\left[\sum_{i=1}^k a_i x_i\right] f(x_1, \dots, x_k) \frac{k}{\pi} dx_i$$

Integrate consecutively w.r.t. $x_k, x_{k-1}, \dots, x_1$ and used the beta integral

$$B(m, n) = \int_0^1 z^{m-1} (1-z)^{n-1} dz.$$

It follows that

$$M(a_1, \dots, a_k) = \sum_{l_1=0}^{\infty} \sum_{l_2=0}^{\infty} \dots \sum_{l_{k-1}=0}^{\infty} \frac{(n_1)_{l_1} (n_2)_{l_2} \dots (n_{k-1})_{l_{k-1}}}{(m-l_1-l_2-\dots-l_{k-1})_{l_1+l_2+\dots+l_{k-1}}} \cdot \\ \cdot \frac{[a_1(1-a_k)]^{l_1}}{l_1!} \frac{[a_2(1-a_k)]^{l_2}}{l_2!} \dots \frac{[a_{k-1}(1-a_k)]^{l_{k-1}}}{l_{k-1}!} \\ = \mathbb{H}(n_1, n_2, \dots, n_{k-1}, m-l_1-l_2-\dots-l_{k-1}; a_1(1-a_k), a_2(1-a_k), \dots, a_{k-1}(1-a_k))$$

as follows from Erdélyi et al (1953) p. 226.

The m.g.f. of (3.3) is defined by

$$M(a_{(1)}, a_{(2)}) = \iint \exp[\operatorname{tr}(a_{(1)} x_{(1)} + a_{(2)} x_{(2)})] \frac{1}{\Gamma_p(n) \Gamma_p(m)}$$

$$\cdot \exp[-\text{tr } x_{(2)}(I + x_{(1)})] |x_{(2)}|^{n+m-\frac{1}{2}(p+1)} |x_{(1)}|^{n-\frac{1}{2}(p+1)} dx_{(1)} dx_{(2)}$$

Integrate $x_{(2)}$ by using the integral

$$\begin{aligned} \int_{x_{(2)} > 0} \exp[-\text{tr } x_{(2)}(I + x_{(1)} - a_{(2)})] |x_{(2)}|^{n+m-\frac{1}{2}(p+1)} dx_{(2)} \\ = \Gamma_p(n+m) |I + x_{(1)} - a_{(2)}|^{-(n+m)} \end{aligned}$$

as derived by Constantine (1963).

Using the transformation (de Waal (1968))

$$y_{(1)} = (I - a_{(2)})^{-\frac{1}{2}} x_{(1)} (I - a_{(2)})^{-\frac{1}{2}}$$

where the Jacobian is $J(x_{(1)} \rightarrow y_{(1)}) = |I - a_{(2)}|^{\frac{1}{2}(p+1)}$
and integrating $y_{(1)}$ by using the integral

$$\int_{y_{(1)} > 0} |y_{(1)}|^{n-\frac{1}{2}(p+1)} |I + y_{(1)}|^{-(n+m)} \exp[\text{tr } a_{(1)}(I - a_{(2)})^{\frac{1}{2}} y_{(1)} (I - a_{(2)})^{\frac{1}{2}}] dy_{(1)}$$

$$= \Gamma_p(n, K) \sum_{k=0}^{\infty} \sum_K (n+m)_K \frac{C_K(a_{(1)}(I - a_{(2)})^{-1})}{|I - a_{(1)}(I - a_{(2)})|^{-n}}$$

$$\text{where } (n+m)_K = \frac{\Gamma_p(n+m, K)}{\Gamma_p(n+m)}$$

and $C_K(T)$ is a zonal polynomial, as derived by Constantine (1963), it follows that

$$\begin{aligned} M(a_{(1)}, a_{(2)}) &= \frac{\Gamma_p(n+m)}{\Gamma_p(m)} |I - a_{(2)}|^{-(n+m)} |I - a_{(1)}|^{-n} \\ &\cdot \sum_{k=0}^{\infty} \sum_K (n)_K (n+m)_K C_K(a_{(1)}(I - a_{(2)})^{-1}). \end{aligned}$$

Further properties of these distributions follow clearly.

The author wishes to thank Prof. H.S. Steyn for his valuable suggestions.

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Manuscript received 14th July 1970.